作業3

1. 證明：$\nabla\times\left(\frac{\vec m\times\vec R}{R^3}\right) = -\nabla\left(\frac{\vec m\cdot\vec R}{R^3}\right)$

左式 $= -\nabla\times\left[\vec m\times\nabla\left(\frac1R\right)\right]$

$= -\underbrace{\left(\nabla\cdot\nabla\left(\frac1R\right)\right)}_{0}\vec m + (\vec m\cdot\nabla)\nabla\left(\frac1R\right) - \underbrace{\left(\nabla\left(\frac1R\right)\cdot\nabla\right)\vec m}_{0} + \underbrace{(\nabla\cdot\vec m)\nabla\left(\frac1R\right)}_{0}$

$= -(\vec m\cdot\nabla)\frac{\vec R}{R^3} = -\left(\nabla\left(\frac{\vec m\cdot\vec R}{R^3}\right) - \frac{\vec R}{R^3}\cdot\underbrace{\nabla\vec m}_{0}\right) = -\nabla\left(\frac{\vec m\cdot\vec R}{R^3}\right)$

2. $\begin{cases}\nabla\cdot\vec D = \rho_f = \varepsilon\nabla\cdot\vec E\\ \nabla\cdot\vec P = -\rho_p = \chi_e\varepsilon_0\nabla\cdot\vec E\\ \chi_e+1 = \varepsilon_r\end{cases}$

$\Rightarrow \rho_p = -\chi_e\varepsilon_0\nabla\cdot\vec E = -\chi_e\varepsilon_0\cdot\frac{\rho_f}{\varepsilon} = -\frac{\varepsilon_r-1}{\varepsilon_r}\cdot\rho_f$

$= -\left(1-\frac{1}{\varepsilon_r}\right)\rho_f = -\left(1-\frac{\varepsilon_0}{\varepsilon}\right)\rho_f$

3. (1) 當 $0\le r<r_1$, $E(r)\cdot 4\pi r^2 = 0 \Leftrightarrow \vec E(\vec r)=0$

當 $r_1\le r<r_2$, $D(r)\cdot4\pi r^2 = \rho_f\cdot\frac43\pi(r^3-r_1^3) \Leftrightarrow D(r)=\rho_f\cdot\frac13\left(r-\frac{r_1^3}{r^2}\right)$

$\Leftrightarrow \vec E(\vec r) = \frac{\vec D(\vec r)}{\varepsilon} = \frac{\rho_f}{3\varepsilon}\left(1-\frac{r_1^3}{r^3}\right)\vec r$

當 $r\ge r_2$ 時, $E(r)\cdot4\pi r^2 = \frac{\rho_f\cdot\frac43\pi(r_2^3-r_1^3)}{\varepsilon_0} \Leftrightarrow \vec E(\vec r) = \frac{\rho_f}{3\varepsilon_0}\cdot\frac{r_2^3-r_1^3}{r^3}\vec r$

(2) $\rho_p = -\left(1-\frac{\varepsilon_0}{\varepsilon}\right)\rho_f$

$\sigma_p(\vec r_1) = \left(\vec P_{空心}-\vec P(\vec r_1)\right)\cdot\vec e_n$

$= -\left(1-\frac{\varepsilon_0}{\varepsilon}\right)\vec D(\vec r_1)\cdot\vec e_n$

$= -\left(1-\frac{\varepsilon_0}{\varepsilon}\right)\frac{\rho_f}{3}\left[1-\left(\frac{r_1^3}{r_1^3}\right)\right]r_1 = 0$

$\sigma_p(\vec r_2) = \left(\vec P(\vec r_2)-\vec P_{外}\right)\cdot\vec e_n$

$= \left(1-\frac{\varepsilon_0}{\varepsilon}\right)\frac{\rho_f}{3}\left[1-\left(\frac{r_1^3}{r_2^3}\right)\right]r_2$

4. $\nabla\times\vec H = \vec J_f + \frac{\partial\vec D}{\partial t} = \vec J_f$

(1) 當 $0\le\rho<r_1$ 時, $\vec H(\vec\rho)=0 \Leftrightarrow \vec B(\vec r)=0$

當 $r_1\le\rho<r_2$ 時, $H(\vec\rho)\cdot2\pi r = J_f\cdot\pi(r^2-r_1^2)$

$\Leftrightarrow \vec H(\vec\rho) = \frac{J_f}{2}\left[1-\left(\frac{r_1}{r}\right)^2\right]r\hat e_\theta$

$\Leftrightarrow \vec B(\vec\rho) = \frac{\mu J_f}{2}\left[1-\left(\frac{r_1}{r}\right)^2\right]r\hat e_\theta$

當 $\rho\ge r_2$ 時, $B(\vec\rho)\cdot2\pi r = \mu_0 J_f\,\pi[r_2^2-r_1^2]$

$\Leftrightarrow \vec B(\vec\rho) = \frac{\mu_0 J_f}{2}\cdot\frac{r_2^2-r_1^2}{r}\hat e_\theta$

(2) $\vec M = \frac{\vec B}{\mu_0}-\vec H$

$\vec J_M = \nabla\times\left(\frac{\vec B}{\mu_0}-\vec H\right) = \nabla\times\left(\frac{\mu\vec H}{\mu_0}-\vec H\right) = \left(\frac{\mu}{\mu_0}-1\right)\nabla\times\vec H = \left(\frac{\mu}{\mu_0}-1\right)\vec J_f$

(3) $\vec M = \left(\frac{\mu}{\mu_0}-1\right)\vec H = \left(\frac{1}{\mu_0}-\frac1\mu\right)\vec B$

$\vec\alpha_M(\vec r_1) = 0$

$\vec\alpha_M(\vec r_2) = \vec e_n\times\left[0-\left(\frac{\mu}{\mu_0}-1\right)\cdot\frac{J_f}{2}\left[1-\left(\frac{r_1}{r_2}\right)^2\right]r_2\hat e_\theta\right]$

$= \left(1-\frac{\mu}{\mu_0}\right)\cdot\frac{J_f}{2}\left[1-\left(\frac{r_1}{r_2}\right)^2\right]r_2\hat z$

5. $\vec J_M = \nabla\times\vec M = \nabla\times\left(\frac{\vec B}{\mu_0}-\vec H\right) = \nabla\times\left(\frac{\mu\vec H}{\mu_0}-\vec H\right) = \left(\frac{\mu}{\mu_0}-1\right)\nabla\times\vec H = \left(\frac{\mu}{\mu_0}-1\right)\vec J_f$